

Institute Of Mathematics and Computer Science
University Of Sindh Jamshoro

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<u>ROLL NO:</u>	2K24/CSE/19
<u>Course:</u>	HCI vs Computer Graphics
<u>SUBMITTED TO:</u>	Mr. Rajesh Kumar
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Part 1

Human-Computer Interaction (HCI):

HCI is the way people interact with computers and other digital devices.

It focuses on making computers easy, useful, and comfortable to use.

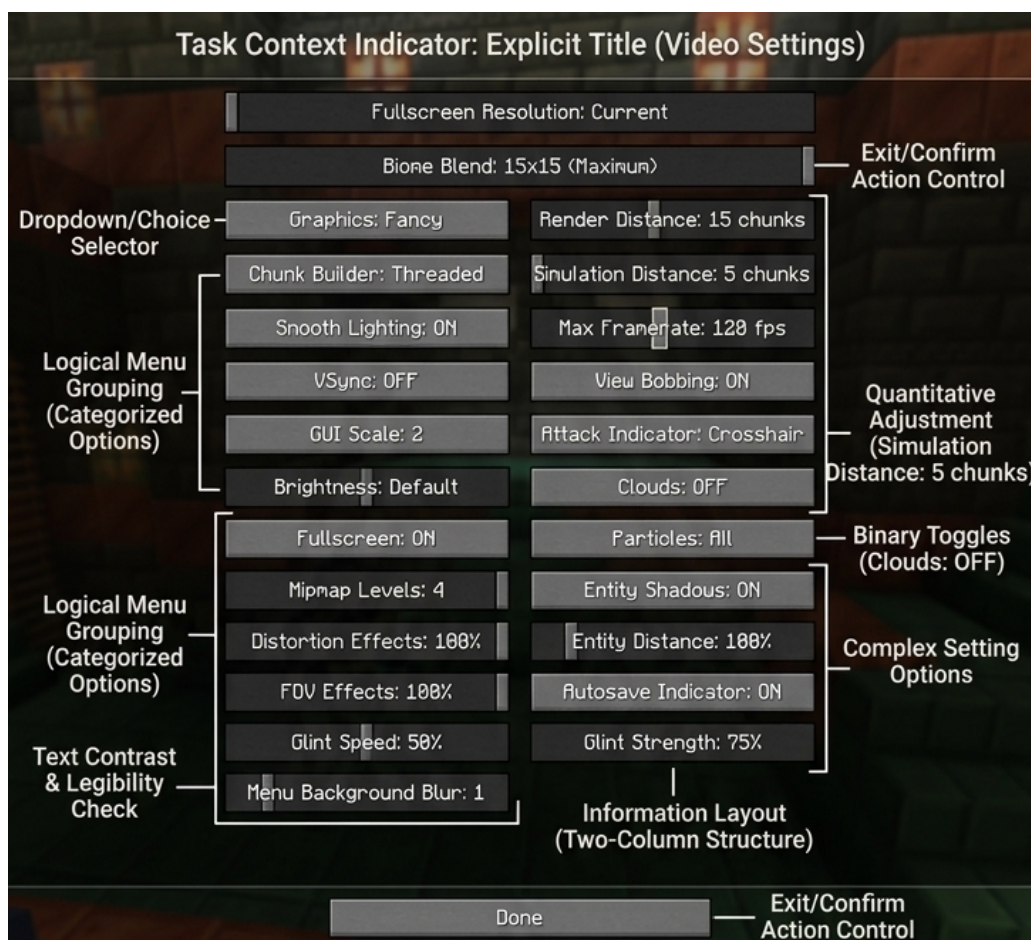
Examples:

- Clicking buttons with a mouse
- Touching icons on a phone
- Typing on a keyboard
- Using voice commands

HCI in Minecraft:

The ways a player can interact with the game:

- Using the mouse to look around and break/place blocks.
- Using the keyboard (WASD) to move.
- Opening the inventory and dragging items.
- Clicking buttons in the Minecraft menu.
- Using a controller to move and interact.
- Changing settings through the game menu



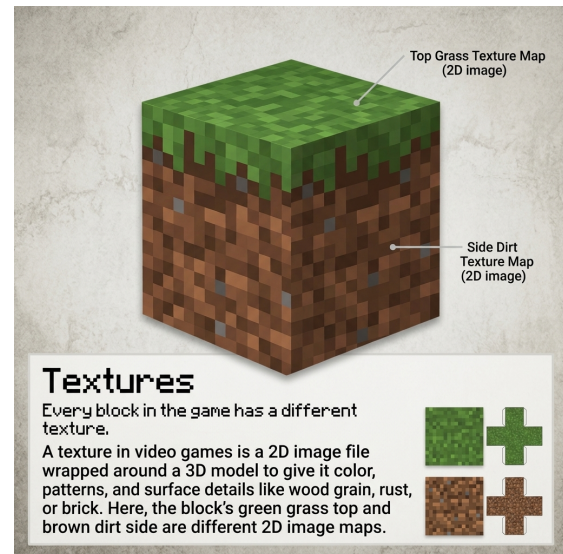
Part 2

Computer Graphics (CG):

Computer Graphics is the creation and manipulation of images, drawings, animations, and visual objects using computers.

Examples:

- 3D models
- Movies and animations
- Computer-generated images
- Textures of the blocks
- Grass, stone, and other blocks
- Trees, animals, and mobs
- Sun, moon, clouds, and sky
- Water and lava
- Lighting, shadows, and particles
- Buildings and structures



Part 3

Question 1: How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

Good computer graphics try to balance visual quality, ease of use, and performance.

Realistic Lighting: Realistic lighting helps users understand depth and see the difference between surfaces, making it easier to move around a 3D environment. However, too much shadow or brightness can hide important details. High-quality lighting can also require more processing power, which may cause lag or lower FPS.

3D View: A 3D view allows users to look at an object from different angles and better understand its shape, size, and position. However, constantly controlling the camera can be difficult and may make the interface harder to understand. Some objects may also be hidden from view, and excessive camera movement can cause motion sickness.

Question 2: What is one simple HCI change you would make to improve the user interface?

I would add a *mini-map in the top-right corner of the screen*. This would help the player see where they are and where they are heading instead of wandering around and getting lost.

The mini-map could also show useful information such as the *time of day, current biome, weather, and coordinates*. This would make it easier for the player to understand their surroundings and navigate the game.